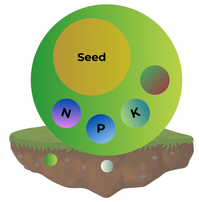
GreenOrbs

Balls that restore forest ecosystems



Fertilizer balls dropped from drones onto specific areas to improve greenery.

Project mission

Creating an innovative, comprehensive ESG solution for businesses and government agencies.
Effective and environmentally friendly land reclamation, providing clients with transparent data on forest growth that can be included in sustainability reports.

Problem analysis

According to the UN Food and Agriculture Organization (FAO), more than 420 million hectares of forest have been lost due to deforestation worldwide, with up to 10 million hectares of forest cover being lost annually between 2015 and 2020.
Kazakhstan has also been affected by this problem: 60,100 hectares of forest cover have been lost due to deforestation (according to Global Forest Watch).

The need to solve the problem

- Forests absorb about 230 billion tons of CO2 annually.
- **Forest cover prevents global warming. For example, tropical forests reduce the average global temperature by 1-2°C.**
- **Forest loss also negatively impacts flora and fauna, with up to 80% of flora and fauna potentially losing their homes due to this problem.**
- **Without the protective layer of forests, soil can become vulnerable to extreme climate conditions, which can lead to erosion, negatively affecting agriculture.**

Beneficiaries

- ***Government agencies of the Republic of Kazakhstan actively investing in environmental technologies and projects***
- ***Major mining companies. Due to increased environmental controls and increased investment in land reclamation, companies are increasing their spending on reclamation and environmental protection.***

Potential partners

Seed and fertilizer suppliers

- ***AlemAgro is the leading seed supplier in Kazakhstan in 2021.***
- ***KazAzot is a nitrogen fertilizer producer.***
- ***KazPhosphorus is a producer of phosphate fertilizers.***
- ***A.S.K Kazakhstan - fertilizers and plant protection products***
- ***KosAgroKommerts - production of mineral fertilizers and pesticides***

https://eldala.kz/rating/11687-top-50-postavshchikov-vysokokachestvennyh-semyan-v-kazahstane-v-2021-godu

Drone suppliers

KazUAV

Yesil Technology *(Forbes.kz - Chinese company to produce industrial drones in Kazakhstan)*

a new company with great potential and investments of over $12 million

Research institutes

- *Kazakh Research Institute of Agriculture and Plant Growing*
- *Department of Botany, E. Buketov Karaganda State University*
- *Laboratory of the National Institute of Emergency Medicine, Karaganda*

Environmental organizations

- *Association of Environmental Organizations of Kazakhstan*
- *Sustainable Development Fund "Kazyna"*

Social innovation

We effectively combine cutting-edge technologies from around the world and adapt them to local market conditions.

- The balls are scattered by drones whose code is based on artificial intelligence;
- Application of NDVI
- The latest biotechnology is used to identify and implement nutrients;
- Biodegradable materials are used
- Application of hydrogels, biochar, capsaicin

Main activity

The production of balls and the implementation of the process of greening forests and reclaimed areas using drones from the company Monitoring and data provision Plant growth control from balls

Processes

Systematization, adaptation, testing

Receiving feedback from experts

Registration of copyright, patent

Promotion and Marketing

Production and packaging

Logistics and storage

Programming, route setting, drone control

Monitoring and reporting

Resources

Technological equipment

drones, GPS, cameras

Seeds, phytohormones. fertilizers

IT infrastructure

drone control and growth monitoring software, mobile app

Information about fertilizers in seeds

Team

agronomists, biologists, drone pilots, engineers, marketers







Unique value proposition

Rapid restoration of vegetation

The balls can cover large areas in a short time.

An eco-friendly approach

The balls are biodegradable and do not harm the environment. Minimal harm during production

Automated and efficient ecosystem restoration

fewer workers are needed for landscaping

Spot greening

changing the composition of the balls, selecting optimal plants for specific areas

Mobile application for growth monitoring

Customers receive growth data through a mobile offering

GreenOrbs significantly differentiates itself from existing reforestation and seeding solutions by combining biotechnology, automation, and data-driven monitoring into a single, integrated system. GreenOrbs improves existing processes, reducing costs per hectare by 2-4 times and increasing seed survival through a nutrient-enriched, climate-adapted formulation.

Customer Relations

B2G*

Negotiations with representatives of the Ministry of Ecology
State programs and tenders (Kazakhstan Environmental Fund) Reporting and transparency, provision of work documents Long-term contracts for the implementation of environmental projects, landscaping and land reclamation

*B2B

Individual approach, proposals for the reclamation of territories
Demonstrations, test runs, pilot projects for feedback
Warranties and service support, regular reports

B2C*

Online consultations (explanation of benefits, assistance in choosing a landscaping program)
Loyalty and discounts, bonus system, special offers
Tracking capability via mobile app

Channels

Exhibitions and conferences (examples of past ones)

- GCIP Kazakhstan 2024
- International Environmental Forum "Ecology, Green Economy, and Innovation – a Factor in Sustainable Development of the Mangystau Region"
- International Scientific and Environmental Forum

Direct sales

Public procurement and tenders (goszakup.gov.kz, etender.kz)
Negotiations with akimats and environmental organizations

Online sales

Website with pricing and detailed information SEO and content marketing Targeted advertising on social media to attract customers Mobile app with an interactive map

Partner channels

Projects with environmental organizations (AEOK) Corporate cooperation (ESG companies) Grants (international and government)
Major investors in the SDG sector

Clients

B2G

Target audience

The Ministry of Ecology, the Ministry of Agriculture, regional administrative entities, forestry enterprises and nature reserves

The purpose of cooperation

Restoring forests after deforestation, fires, and drought (in particular, fires in the North Kazakhstan region, Kostanay, and Abay regions); combating desertification, and establishing protective forest belts along fields and highways to achieve sustainable development.

Volume

State greening programs allocate millions of tenge for ecosystem restoration, with 21.3 billion tenge allocated in 2023.

B2B

Target audience

Mining and metallurgy, oil and gas, construction companies, agricultural holdings, farms

The purpose of cooperation

Compensation for environmental damage, reduction of carbon footprint, reclamation of quarries, mines, waste heaps, increasing the productivity of agricultural land, achieving the SDGs in the field of ecology, fulfilling obligations

Volume

In 2022, large companies in Kazakhstan spent more than 15 billion tenge on reclamation. “ERG” is an international group of companies, whose branch in Kazakhstan specializes in landscaping in Karaganda (4,993 seedlings), Pavlodar (1,880,000)

B2C

Target audience

Owners of country houses, dachas, environmental activists, small agricultural enterprises

The purpose of cooperation

Greening personal plots without wasting time and effort, combating erosion on farmland, supporting environmental initiatives through funding

Volume

The average bill can be 50,000-300,000 tenge for a package of services. Potential audience: tens of thousands of people across the country.

B2G - business for the government
B2B - business for companies *B2C - business for private clients

Estimated income

Price for services

№	Service/product	Price, KZT
1.	B2G. Basic landscaping	800,000 / ha
2.	B2B. Corporate ESG and Industrial Land Reclamation	1,500,000 / year
3.	B2B. Premium Analytics and Guarantees	from 2,000,000 / year
4.	B2C. Greening of suburban areas	100,000 - 300,000 / order
5.	B2C. Planting trees named after environmental campaigners.	50,000 / order

Income forecasting

Сегмент	1-ый год	2-ой год	3-й год	4-ый год	5-ый год
B2B					
Количество гектар	6	20	30	40	60
Средняя цена за га	1 500 000	1 500 000	1 600 000	1 600 000	1 600 000
Доход B2B	9 000 000	30 000 000	48 000 000	64 000 000	96 000 000
B2G					
Количество гектар	0	0	10	32.5	48.75
Средняя цена за гектар	-	-	*	800 000	800 000
Доход B2G	0	0	8 000 000	26 000 000	39 000 000
Общий доход	9 000 000	30 000 000	56 000 000	90 000 000	135 000 000

*We plan to focus more on the B2B segment and position ourselves as a high-tech provider of ESG solutions.

*A pricing policy for private clients was also developed. However, due to inefficiencies, the need for significant marketing investments, and low profitability, GreenOrbs will not target B2C customers in the first few years.

QR code for viewing detailed comments on expenses and income

Estimated costs

Initial (capital) costs (CAPEX)

No. s/n	Наименование	Количество	Цена, KZT	Общая сумма, KZT
1.	Промышленный/ агрокультурный дрон	1	4 000 000	4 000 000
2.	Оборудование для производтсва шариков (pilot line)	-	10 000 000	10 000 000
3.	Разработка платформ (моб.приложение и сайт)	-	3 000 000	3 000 000
4.	Лицензии и разрешения	-	249 116	249 116
5.	Исходные материалы для создания прототипов	-	500 000	500 000
6.	Резервный фонд (15% от всего бюджета)	-	2 662 367	2 662 367
Итого				**20 411 483**

Operating expenses for the first year (OPEX)

No.	Наименование	Количество	Цена, KZT	Общая сумма, KZT
1.	Стоимость производства шариков (1.8 тонн)	-	5 447 700	5 447 700
2.	Зарплаты сотрудников и налоговые отчисления	-	12 960 000	12 960 000
3.	Маркетинг и продажи	-	1 500 000	1 500 000
4.	Коммунальные услуги, программное обеспечение, связь	-	1 000 000	1 000 000
5.	Поддержка оборудования	-	500 000	500 000
6.	Амортизация оборудования	-	2 882 000	2 882 000
Итого				**24 290 000**

Annual costs: Information provided is based on the project being in its early stages of development.

***We add a reserve fund to our expenses, which is 15% of the remaining costs. According to our calculations, this amount is 2,662,367 tenge. A reserve fund is standard practice for covering financial risks and expenses.**

The cost of 1 kg of the composition for the balls is 3026.5 tenge/kg

QR code for viewing detailed comments on expenses and income

Financial result

Индикатор	Год развития проекта				
	В год получения инвестиций и грантов	Следующий год	На 3-й год	4-й год	5-й год (достижение точки возврата инвестиций)
Доходы					
B2B	9 000 000	30 000 000	48 000 000	64 000 000	96 000 000
B2G	0	0	8 000 000	26 000 000	39 000 000
Общий доход	9 000 000	30 000 000	56 000 000	90 000 000	135 000 000
Расходы					
Себестоимость производства шариков	5 448 000	18 159 000	33 899 000	54 477 000	81 716 000
Операционные расходы(без учета расходов на шарики)	15 960 000	16 918 000	17 933 000	19 009 000	20 149 000
Амортизация капитальных расходов	2 882 000	2 882 000	2 882 000	2 882 000	2 882 000
Общие расходы	24 290 000	37 959 000	54 714 000	76 368 000	104 747 000
Итого					
Общая прибыль	-15 290 000	-7 959 000	1 286 000	13 632 000	30 253 000

NPV calculation

Year	Income	Expenses	Net profit	Current Value of Earnings
1	9 000 000	24 290 000	-15 290 000	–12 958 000
2	30 000 000	37 959 000	-7959000	–5 716 000
3	56 000 000	54 714 000	1 286 000	783 000
4	90 000 000	76 368 000	13 632 000	7 033 000
5	135 000 000	104 747 000	30 253 000	13 230 000

Net present value = – 18,674,000 + 21,046,000 = 2,372,000 tenge Discount rate 18%, positive NPV in the 5th year of project development

For the first year:

Cost of 1 hectare = Cost of production / Number of hectares = 5,447,700 / 6 = 908,000 tenge

Gross Profit = Price - Cost = 1,500,000 - 908,000 = 592,000 tenge/ha ~ 39% (592,000 / 1,500,000) Net Profit = -15,290,000 / 9,000,000 = -170% (due to high initial costs and small scale)

Break-even point = fixed costs per year / margin = 16,000,000 / 592,000 ~ 27 hectares

For the following years:

The cost of landscaping per hectare will fall due to the effect of scale.

Net margin will become positive in year 3 (~2%) and will grow continuously due to economies of scale.
For example: net margin in year 5 will be ~22% (30.2 / 135 million).

The initial investment will be recouped in 5 years.

Project impact

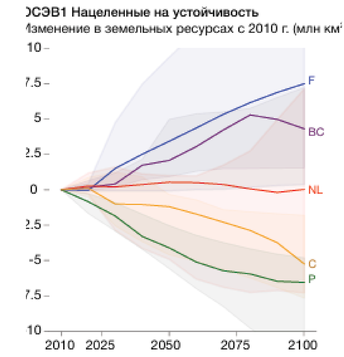
New jobs

The production, distribution and planting of seed balls creates new jobs, contributing to the improvement of the economy and the standard of living. The possibility of cooperation with regions with high unemployment.

Impact on the environment

Helps in forest restoration, recovery and growth.
In 5 years
~200–250 hectares of restored land
Potential absorption of 2,000–5,000 tons of CO_2 annually
Enhancing biodiversity in drylands

***Increase in forest area by 7.5 million km²**
will reduce the average temperature of the Earth by 1.5°C by 2100



Change in forest numbers (blue), ***bioenergy crops (purple), natural lands (red), arable and pasture lands (yellow and green)***

Sustainable Future

promotes the development of a green economy and attracts investment in projects aimed at a green future, motivating people to participate in environmental initiatives

Improving air quality

Urban trees can reduce fine particulate matter (PM2.5) levels by 7–24%, removing approximately 711,000 tons of pollutants from urban air.

Contribution to the economy

Reducing the cost of reclamation by 20–30% compared to traditional methods; Possibility of scaling without a proportional increase in costs; Support for ESG strategies of large companies

Project profitability

Why should investors and beneficiaries trust our project?

Constantly growing demand for the product and a wide target audience

The project's high long-term relevance increases demand for Greenorbs in both domestic and international markets, which will continue to grow, as solving the environmental problem is impossible in the short term.

Economic analysis

The conducted economic analysis shows the competitiveness of the project and the absence of exact analogues on the market, which makes the implementation cost-effective

Financial analysis

The project's implementation requires capital investment and additional costs, but investments, grants, and sales profits over the coming years will cover all costs.